Symmetry and Decoherence in Quantum Computers

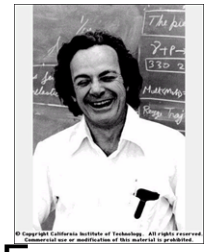
Qualifying Exam Talk
March, 2001

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Outline

- Motivation
- Quantum error correction
- Decoherence and symmetry
- Decoherence-free subspaces and subsystems
- Fault-tolerant Quantum Computation
- Fungible quantum information
- Conclusion

Information is Physical + Physics is Quantum = Quantum Computation


Feynman


Turing

Quantum information processor (unbeautified model)

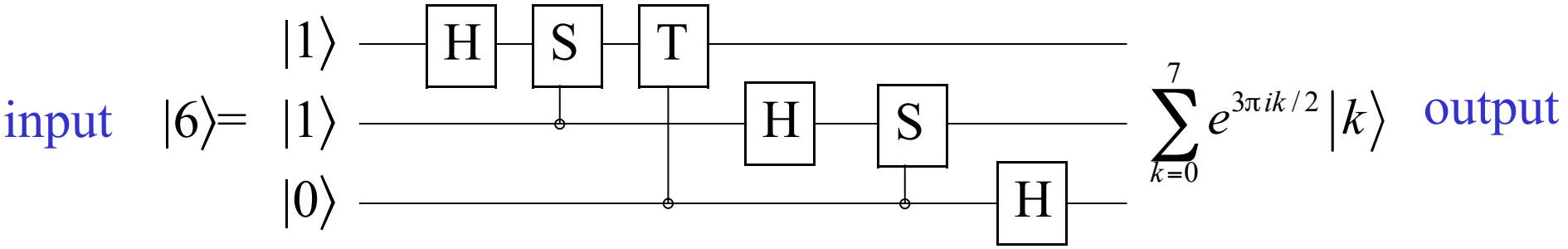
information: states of multiple quantum systems

qubit: $|0\rangle$, $|1\rangle$: pure state is $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$

qubits: pure state is $|\psi\rangle=\Sigma k c_k|k\rangle$ ← n qubits = 2n basis states

manipulation of information: 1. unitary action (gates)

2. measurements



Quantum Fourier Transform on 3 qubits

Why Quantum Compute?

- Quantum computers can compute *algorithmically faster*

  (provable vs classical computation relative to an oracle)

  Grover's search algorithm, Shor's factoring algorithm

- Distributed computation requires *less communication* vs classical

  two parties (one with x, one with y) wish to compute f(x,y)

  # bits of communication required is less in quantum world

- *Efficient simulation of quantum systems*

  fermions, bosons, topological quantum field theories, ….

- Quantum cryptography: provably *secure key distribution*

Sample Quantum Communication Complexity

B: y0y1

A: x0x1

C: z0z1

Three parties A, B, C given inputs x,y,z

Want to compute f(x,y,z) via a set protocol of communication.

Ability to "broadcast" information to other two parties. cost=# bits broadcast

SAMPLE WHERE PRESHARED ENTANGLEMENT LOWERS COST

A, B, C each given a two bit string.

guarantee: x0y0z0∈{000, 011, 101, 110}, x1y1z1 unrestricted

f(x,y,z)= x1⊕y1⊕z1⊕(x0∨y0∨z0) (⊕ is XOR, ∨ is OR)

Quantum: each party has one part of a tripartite entangled state:

$$|\psi\rangle = \tfrac{1}{2}\left(|000\rangle - |011\rangle - |101\rangle - |110\rangle\right)$$

$$|abc\rangle = |a\rangle \otimes |b\rangle \otimes |c\rangle$$

A B C

Protocol:

$$\text{Hadamard} \equiv H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

1. For each given party, if first bit (x0,y0, or z0) is 1, then apply the Hadamard gate to given part of $|\psi\rangle$

2. Next, measure the respective qubit in the Pauli Z basis. Denote the given parties output as a,b,c respectively.

If x0y0z0=000, then $|\psi\rangle$ unchanged, a⊕b⊕c=0

If x0y0z0=110, then $|\psi\rangle \rightarrow \frac{1}{2}\left(|010\rangle - |001\rangle + |111\rangle + |100\rangle\right)$, a⊕b⊕c=1 (etc)

a⊕b⊕c= x0∨y0∨z0

3. Parties broadcast- A: (x1⊕a) B: (y1⊕b) C: (z1⊕c)

Each party can now compute

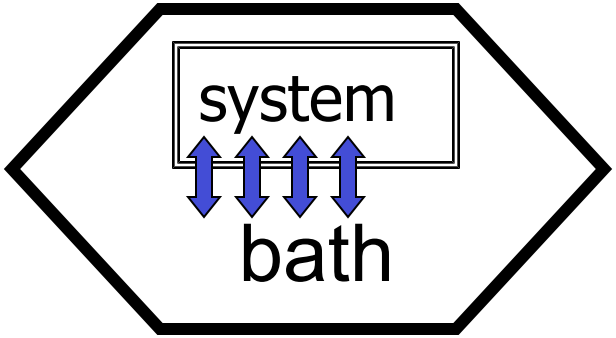
(x1⊕a)⊕(y1⊕b)⊕(z1⊕c)= x1⊕y1⊕z1⊕(x0∨y0∨z0)

f(x,y,z) with 3 bits classical result requires: 4 bits

Buhrman, Cleve, Tapp 1997

Great Benefits! Why Not Quantum Compute?

Real quantum systems are open quantum systems!



full Hamiltonian:

System evolution Bath evolution **Interaction causing DECOHERENCE**

$$H = H_S \otimes I_B + I_S \otimes H_B + \sum_\alpha S_\alpha \otimes B_\alpha$$

← Expansion into system and bath operators

Systems pure states become entangled with environment:

$$|\psi_0\rangle = |\psi\rangle_S \otimes |\psi\rangle_B \rightarrow \sum_n c_n |n\rangle \otimes |n\rangle \qquad \rho_0 = |\psi_0\rangle_S \, {}_S\langle\psi_0| \rightarrow \rho_S \quad Tr[\rho_S] < 1$$

System evolution (by itself) is no longer unitary

Coherent properties of system's quantum information lost

Is Decoherence Really a Problem?

Liberal Estimates (some fudges)

2 level system (qubit)	Switch time T [sec]	Decoherence time τ (upper bound)	Quality factor τ/T (no. of operations)
Exciton in Quantum dot	10^{-15} (single qubit rotation only)	10^{-12}	10^{3}
Electron spin in quantum dot	10^{-12}	10^{-9}	10^{3}
Trapped ion (In)	10^{-6}	10^{-1}	10^{5}
Nuclear spin	10^{-3}	10^{4}	10^{7}
Atom in optical cavity	10^{-14}	10^{-5}	10^{9}
Classical bit	10^{-9}	10^{7}	10^{16}

!! → ← !!

Decoherence problem

not much can be done before quantum information is destroyed

Encoding

Quantum information can be encoded, for example

$$|0_L\rangle \rightarrow \frac{1}{2\sqrt{2}}\left[|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle\right]$$

$$|1_L\rangle \rightarrow \frac{1}{2\sqrt{2}}\left[|000\rangle - |001\rangle - |010\rangle + |011\rangle - |100\rangle + |101\rangle + |110\rangle - |111\rangle\right]$$

Quantum information can now be stored as $|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$

Single phase error $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ can now be fixed!

$$Z \otimes I \otimes I|0_L\rangle \rightarrow \frac{1}{2\sqrt{2}}\left[|000\rangle + |001\rangle + |010\rangle + |011\rangle - |100\rangle - |101\rangle - |110\rangle - |111\rangle\right]$$

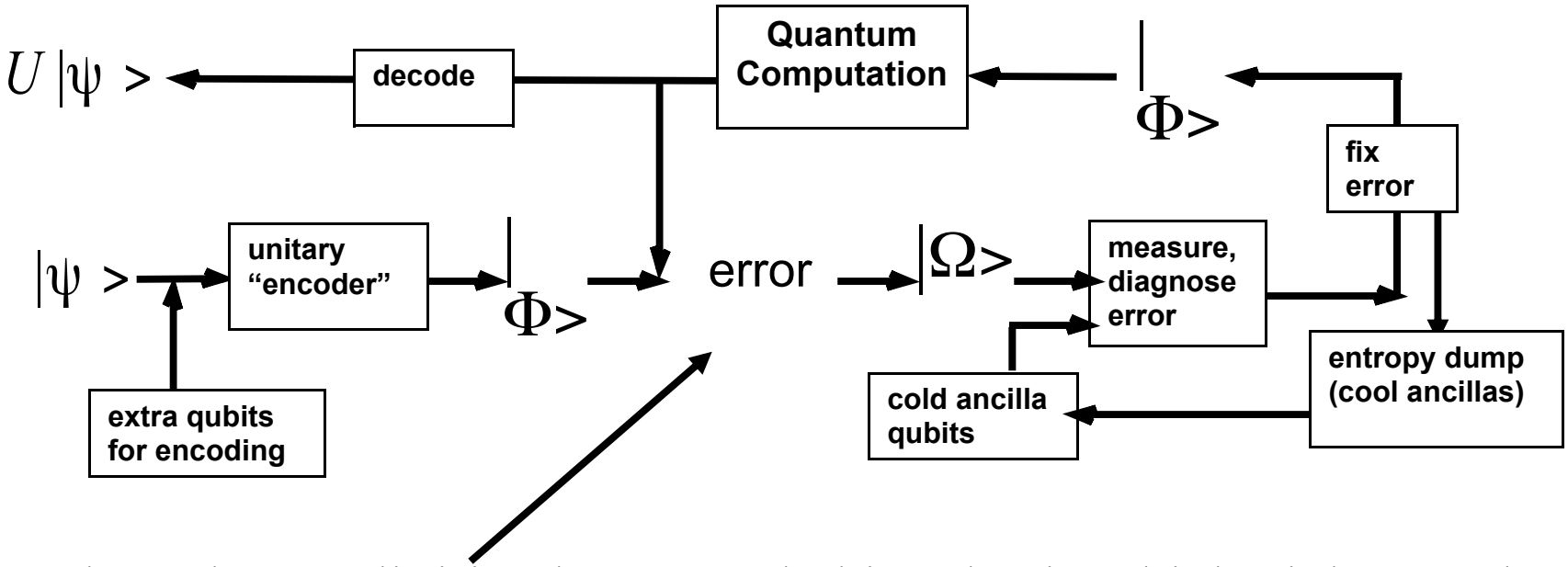
$$Z \otimes I \otimes I|1_L\rangle \rightarrow \frac{1}{2\sqrt{2}}\left[|000\rangle - |001\rangle - |010\rangle + |011\rangle + |100\rangle - |101\rangle - |110\rangle + |111\rangle\right]$$

can be distinguished from

$$I \otimes Z \otimes I|0_L\rangle \rightarrow \frac{1}{2\sqrt{2}}\left[|000\rangle + |001\rangle - |010\rangle - |011\rangle + |100\rangle + |101\rangle - |110\rangle - |111\rangle\right]$$

$$I \otimes Z \otimes I|1_L\rangle \rightarrow \frac{1}{2\sqrt{2}}\left[|000\rangle - |001\rangle + |010\rangle - |011\rangle - |100\rangle + |101\rangle - |110\rangle + |111\rangle\right]$$

Quantum Error Correction



errors have been "digitized": expanded in a basis which picks out the "most likely" errors

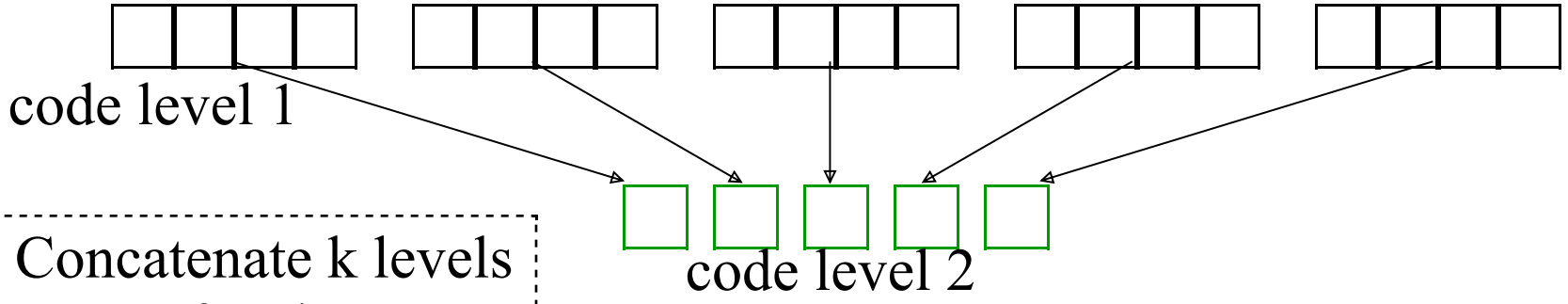
independent decoherence: qubits couple to independent baths

$\langle i | E_\alpha^\dagger E_\beta | j \rangle = \delta_{i,j}$ $\quad |i\rangle \equiv$ encoded state $\quad E_\beta \equiv$ error operator

errors take codewords to orthogonal subspace (diagnosable!)

$\langle a | (XII)(IXI) | b \rangle = 0 \qquad |a,b\rangle \in \{|000\rangle, |111\rangle\}$

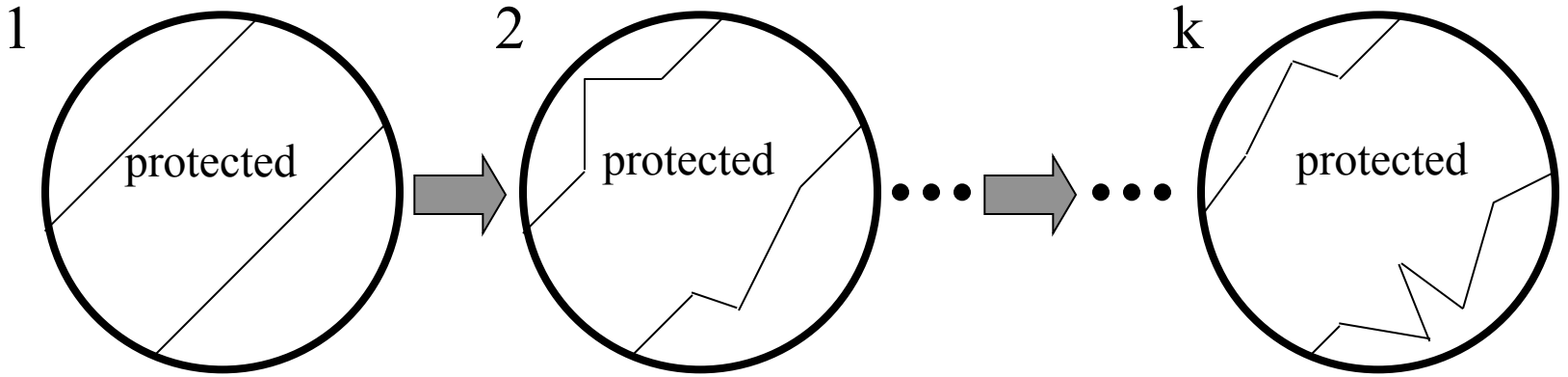
Threshold for Quantum Computation



Concatenate k levels of codes.

Error rate p → cp2 at each level of concatenation.

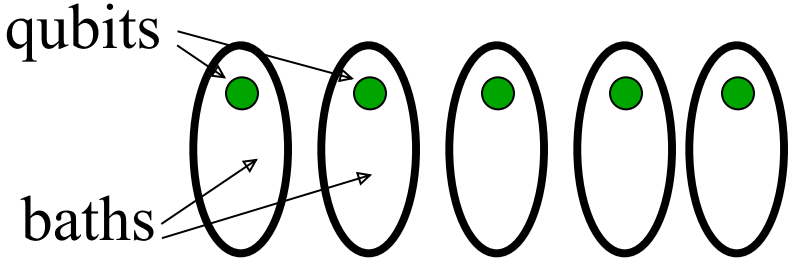
Fail with probability (cp)2k/c. Can make arbitrarily small if p<1/c=pthreshold



error rate competes with *volume of protected states*

Threshold Theorem: if quality factor is > 104 to 106, no decoherence

Decoherence Models



Threshold theorem holds for "independent decoherence"

Correlated errors are "small" (at least "smaller" as # qubits increases)

How generic?

as spacing between qubits gets smaller:

independent environments??

coupling of qubits to a field:

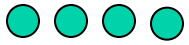
if qubit space << wavelength

then field cannot distinguish qubits

CAN DIFFERENT DECOHERENCE MECHANISM BE EXPLOITED FOR QUANTUM COMPUTATION?

Collective Decoherence

spins coupled to bath field, N qubits $s_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $s_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $s_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$



system-bath Hamiltonian

Bath mode with a wavelength much longer than the inner qubit spacing

$$H_I = \sum_{i=1}^{n} \sum_{\alpha=\{x,y,z\}} c_{\alpha,i} \; s_\alpha^{(i)} \otimes b_{\alpha,i} \quad \Leftrightarrow \quad H_I = \sum_{\alpha=\{x,y,z\}} \sum_{i=1}^{n} c_\alpha s_\alpha^{(i)} \otimes b_\alpha = \sum_{\alpha=\{x,y,z\}} S_\alpha^n \otimes B_\alpha$$

$s_\alpha^{(i)}$: system operator acting on ith qubit

$b_{\alpha,i}$: bath operator

$c_{\alpha,i} = c_\alpha$

$$s_\alpha^{(i)} = I \otimes \ldots \otimes I \otimes \underbrace{s_\alpha}_{i\text{th qubit}} \otimes I \otimes \ldots \otimes I$$

bath couples identically to each qubit

collective spin operators:

$$S_\alpha^n = \sum_{i=1}^{n} s_\alpha^{(i)}$$

$$[s_\alpha, s_\beta] = i\varepsilon_{\alpha\beta\gamma} s_\gamma$$

su(2) Lie algebras

$$[S_\alpha^n, S_\beta^n] = i\varepsilon_{\alpha\beta\gamma} S_\gamma^n$$

Just Angular Momentum Addition

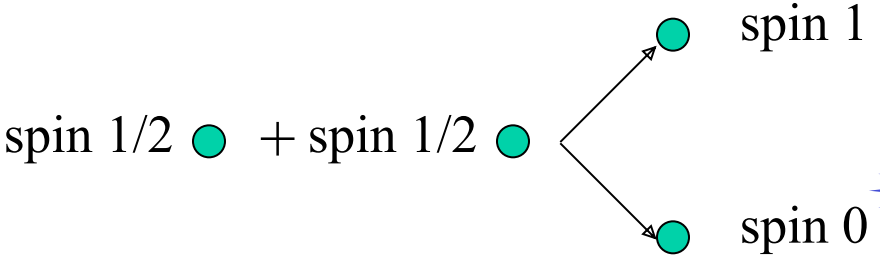
Collective su(2) for n=2 qubits:

$$S_x^2 = \frac{1}{\sqrt{2}}\left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \qquad S_y^2 = \frac{1}{\sqrt{2}}\left(\begin{array}{ccc|c} 0 & -i & 0 & 0 \\ i & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \qquad S_z^2 = \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

triplet states $\begin{cases} |J=1, m_J=1\rangle = |00\rangle \\ |J=1, m_J=0\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) \\ |J=1, m_J=-1\rangle = |11\rangle \end{cases}$ singlet state $\left\{ |J=0, m_J=0\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right) \right.$

red is the 3 dimensional irreducible representation of su(2) (J=1)

blue is the 1 dimensional irreducible representation of su(2) (J=0)



simple addition of angular momentum…

nothing to see here, but:

Degeneracy

Something special happens with n=3:

•spin-1 (triplet) plus spin-1/2 can produce either a spin-1/2 or a spin-3/2.

•spin-0 (singlet) plus spin-1/2 can produce spin 1/2.

$$S_x^3 = \left(\begin{array}{cccc|cc|cc} 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{\frac{3}{2}} & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \end{array}\right)$$

$$\left\{\begin{array}{l} \left|J=\tfrac{3}{2},m_J=\tfrac{3}{2}\right\rangle = \left|000\right\rangle \\ \left|J=\tfrac{3}{2},m_J=\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{3}}\left(\left|001\right\rangle+\left|010\right\rangle+\left|100\right\rangle\right) \\ \left|J=\tfrac{3}{2},m_J=-\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{3}}\left(\left|011\right\rangle+\left|101\right\rangle+\left|110\right\rangle\right) \\ \left|J=\tfrac{3}{2},m_J=-\tfrac{3}{2}\right\rangle = \left|111\right\rangle \end{array}\right.$$

$$\left\{\begin{array}{l} \left|J=\tfrac{1}{2},m_J=\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{2}}\left(\left|01\right\rangle-\left|10\right\rangle\right)\left|0\right\rangle \\ \left|J=\tfrac{1}{2},m_J=-\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{2}}\left(\left|01\right\rangle-\left|10\right\rangle\right)\left|1\right\rangle \end{array}\right.$$

$$\left\{\begin{array}{l} \left|J=\tfrac{1}{2},m_J=\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\left|00\right\rangle\left|1\right\rangle-\sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(\left|01\right\rangle+\left|10\right\rangle\right)\right\}\left|0\right\rangle \\ \left|J=\tfrac{1}{2},m_J=-\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\left|11\right\rangle\left|0\right\rangle-\sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(\left|01\right\rangle+\left|10\right\rangle\right)\right\}\left|1\right\rangle \end{array}\right.$$

These spin-1/2 states come from combining a spin-0 two qubit state and a spin-1/2 single qubit state

These spin-1/2 states come from combining a spin-1 two qubit state and a spin-1/2 single qubit state

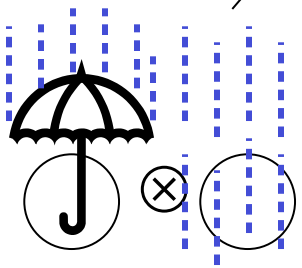
There is a **DEGENERACY** in the action of the 3 qubit collective su(2) operation.

Degeneracy Implies Safety

$$S_\alpha^3 = \left(\begin{array}{c|c:c} s_\alpha^4 & 0 & 0 \\ \hline 0 & s_\alpha^2 & 0 \\ \hdashline 0 & 0 & s_\alpha^2 \end{array}\right) = \left(\begin{array}{c|c} s_\alpha^4 & 0 \\ \hline 0 & I \otimes s_\alpha^2 \end{array}\right)$$

where $s_\alpha^k \equiv$ {the k-dimensional irreducible representation of su(2)}

$$S_\alpha^3 \rightarrow s_\alpha^4 \oplus \left(I_2 \otimes s_\alpha^2\right)$$



protected qubit unprotected qubit

Information encoded into the **DEGENERACY** is **immune** to collective decoherence

(KLV) Emanuel Knill, Raymond Laflamme, Lorenza Viola, Phys. Rev. Lett. **84**, 2525 (2000)

Classical Example

clumsy hands channel: both coins get flipped with probability p

$|HH\rangle$
$|HT\rangle$
$|TH\rangle$
$|TT\rangle$



(1-p)



+p



$|$coin 1, coin 2 $\rangle$

Information can be immune to this channel if

encode into the parity

$|$parity,coin 1$\rangle$

$|1H\rangle=$



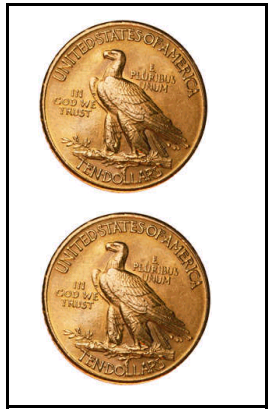
$|1T\rangle=$



$|0H\rangle=$



$|0T\rangle=$



$|$parity$\rangle=|0\rangle$ if coins same, $|$parity$\rangle=|1\rangle$ if coins differ

Decoherence-free Subsystems (KLV)

General system-bath Hamiltonian $H_{SB} = \sum_{\alpha} A_\alpha \otimes B_\alpha$

A_α: system operators; B_α: linearly independent bath operators

A_α form an algebra A (add, multiply them) which is decomposable into subsystems:

$$\mathrm{A} = \bigoplus_{J=1}^{r} \left(I_{n_J} \otimes M_{d_J} \right)$$

M_{d_J}: linear operators on a d_J dimensional space

I_d is the $d \times d$ identity operator

n_J is the degeneracy of the Jth irrep

Quantum information encoded into a single degenerate component subsystem will not decohere! Decoherence-Free Subsystems (DFSs)

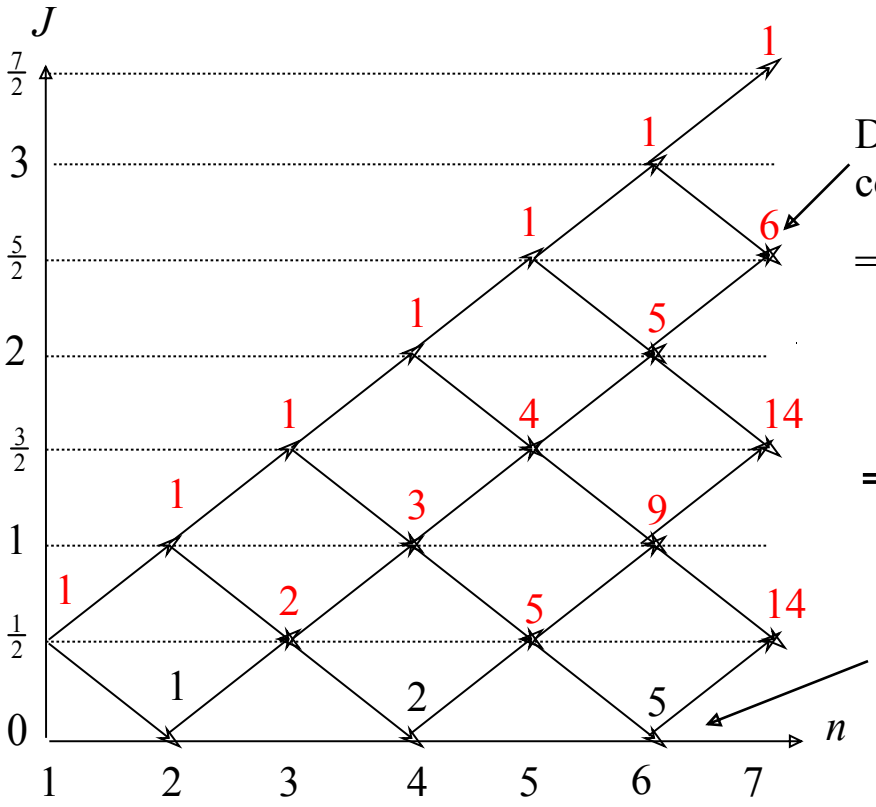
"The universe is an enormous direct product of representations of symmetry groups" - Steve Weinberg

<u>for open system becomes</u>

"The universe is an enormous direct product of full matrix algebras" - not Steve Weinberg

(KLV) Emanuel Knill, Raymond Laflamme, Lorenza Viola, Phys. Rev. Lett. **84**, 2525 (2000)

Collective DFSs



Degeneracy of the n qubit, total angular momentum J collective su(2) operators

= dimension of quantum information which can be encoded into the corresponding DFS.

$$= \frac{(2J+1)n!}{\left(\frac{n}{2}+J+1\right)!\left(\frac{n}{2}-J\right)!}$$

The J=0 irreps of su(2) are 1 dimensional *Decoherence-Free Subspaces*.

$$S_\alpha^n \left|i\right\rangle = 0$$

$$\text{encoding efficiency} \equiv r = \frac{\log_2(\text{dimension of DFS})}{\text{n}}$$

As $n \to \infty$, $r(n) \to 1 - c\frac{\log_2 n}{n}$. The encoding efficiency is nearly unity for large n.

however, physical collective DFS only for small n.

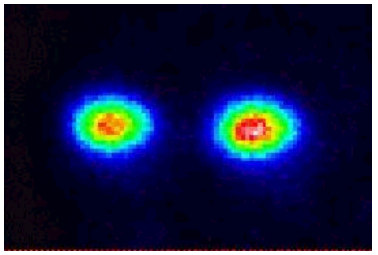
Experimental Demonstration

REPORTS

A Decoherence-Free Quantum Memory Using Trapped Ions

D. Kielpinski,[1]* V. Meyer,[1] M. A. Rowe,[1] C. A. Sackett,[1]
W. M. Itano,[1] C. Monroe,[2] D. J. Wineland[1]

SCIENCE VOL 291 9 FEBRUARY 2001

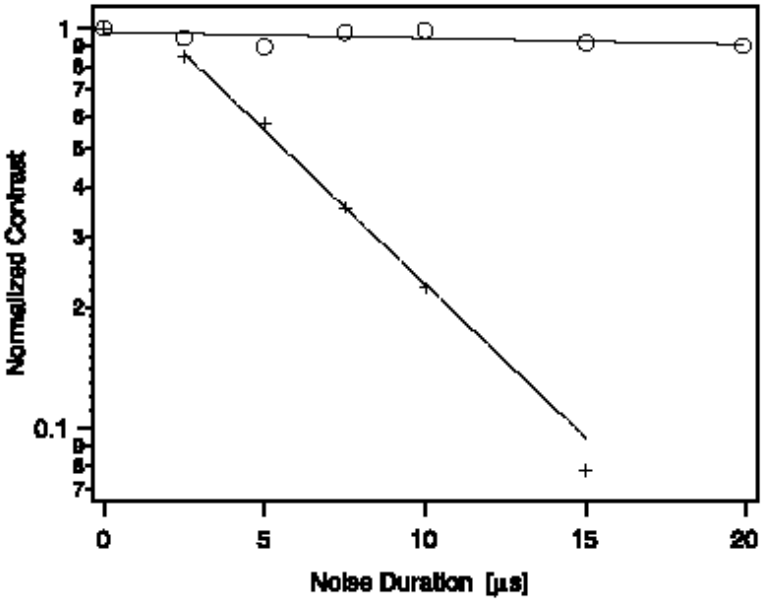


collective phase errors:

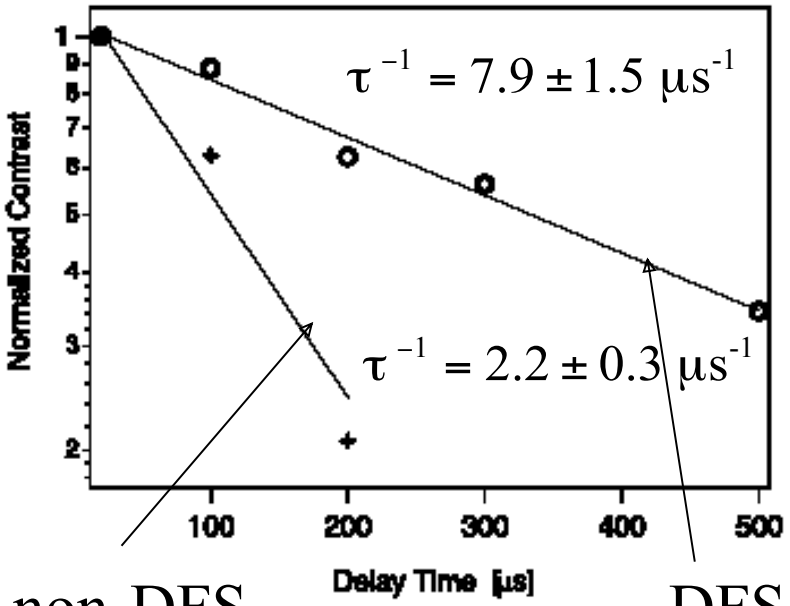
$$S_z^n = \sum_{i=1}^{n} s_z^{(i)}$$

engineered noise (detuned laser)

ambient conditions!

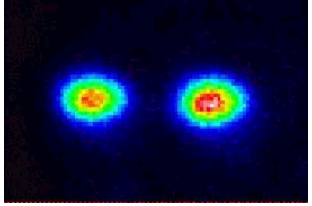


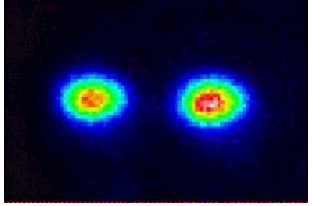
(see also Kwiat et al., Science 2000)

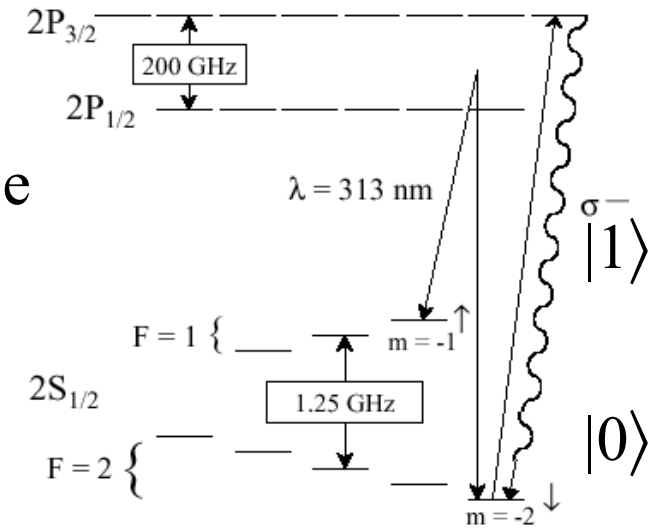


non-DFS DFS

Ion Trap Experiment

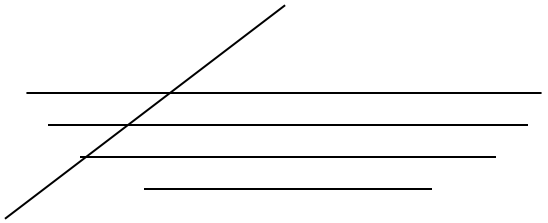




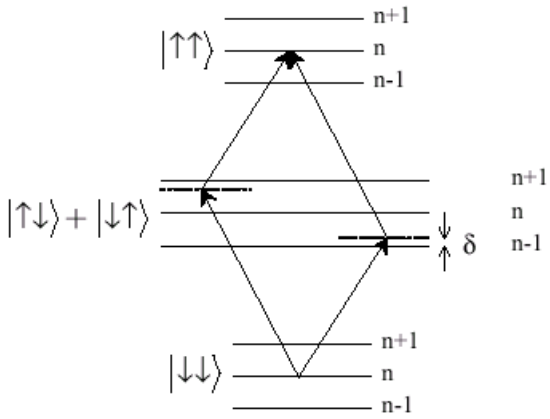


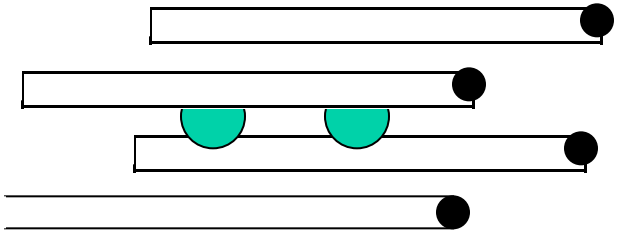
9Be
+

Each qubit is hyperfine level of the ion



center of mass motion used as a bus to couple ions: ~5 MHz





Two stimulated Raman transitions used to take encoding of single ion to two ion encoded states (center of mass acts as bus for communication between ions)

Using DFSs in Quantum Computers

Philosophy:how can the collective DFSs be used to build a quantum computer?

- How to analyze perturbing interactions?
  (Do they make DFSs useless?)
- How to work with error correction?
  (Can DFSs be used in error correcting codes?)
- How can one compute on a DFS?
  (Can leaving a DFS be avoided?)
- How do DFSs fit in with fault-tolerant quantum computation?
  (Fault-tolerant measurements?)

Poking DFSs

perturb a DFS with strength λ:

$$H_{SB} = gH_{DFS} + \lambda H'$$



g >> λ : λg terms in perturbation could be huge?

DFSs are robust to perturbations to first order: λg terms vanish

Lindblad master equation (Markovian approximation)

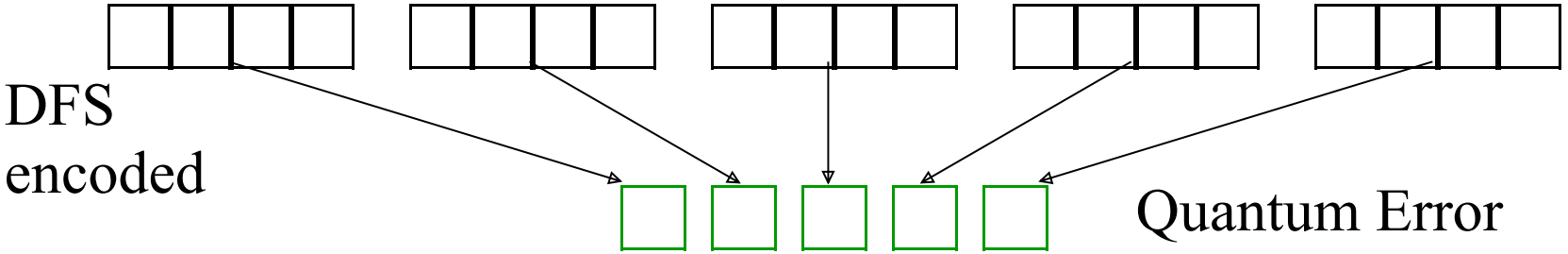
Exact operator sum representation (no markovian approximation)

Lidar, Chuang, Whaley, PRL **81**, 2595 (1998)

Bacon, Lidar, Whaley, PRA **60**, 1944 (1999)

Concatenation of DFS in QECC

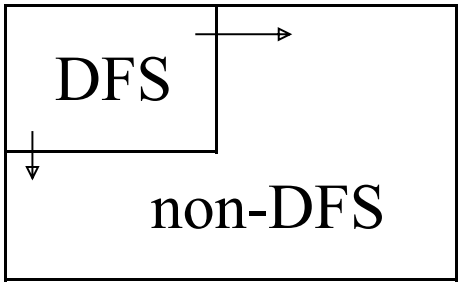
Philosophy: "USE SYMMETRY FIRST!"



Issues:

POKING: OK ✓

- symmetry breaking decoherence couple to symmetric decoherence?
- leakage detection?
- quantum computation on encoded states?



leakage OK in: Lidar, Bacon, Whaley, PRL **82**, 4556 (2000)

Universal Quantum Computation on DFS?

Problem: given a subsystem is it possible to perform computation on the encoded information?

Existential result: always possible

Real world result: using two-body interactions?

respecting geometry?

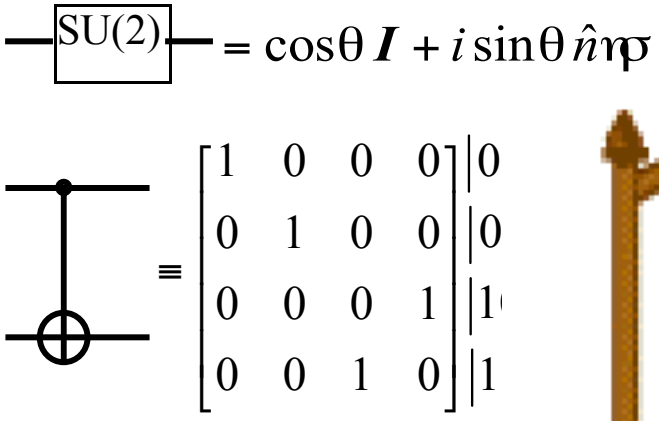
Computation on a subsystem of dimension d

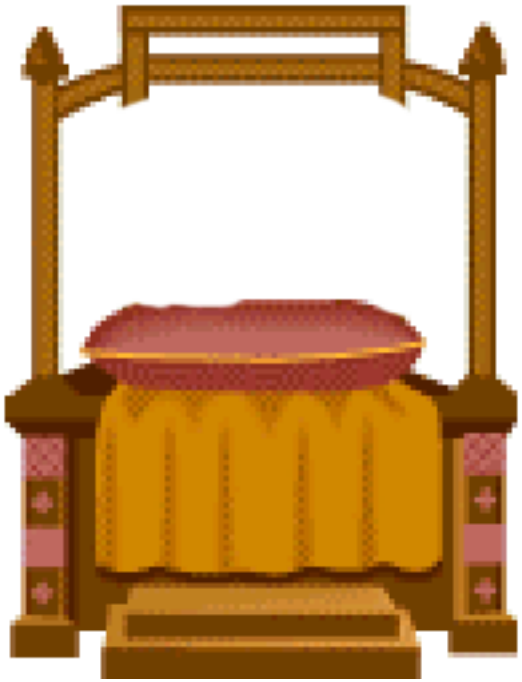
= ability to perform su(d) rotation to good accuracy

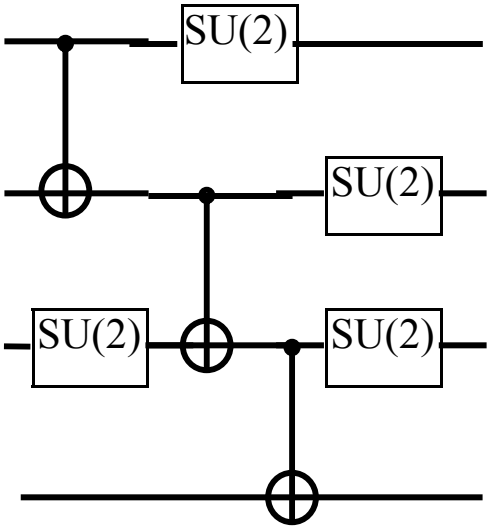
More general question for quantum information:

what interactions do we need to quantum compute?

The Standard







The Royal King and Queen
of Universal Quantum Gates

CNOT and 1-qubit rotations

Combining CNOT and 1-qubit rotations on n qubits gives su(2n)

Exchange's Universality?

The Exchange Hamiltonian on 2 qubits $\equiv E = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$. $E_{ij} \equiv$ E acting between qubits i and j.

$\left[E_{ij}, (S^n)^2\right] = 0$ where $(S^n)^2 = (S_x^n)^2 + (S_y^n)^2 + (S_z^n)^2$ $\quad \left[E_{ij}, S_\alpha^n\right] = 0$

$|J, m_J, \lambda\rangle$

Exchange does not change J or mJ, does mix degeneracy!

Consider the ability to use the exchange Hamiltonian between qubits (nearest neighbors is enough)

$$U_i(t) = \exp\left[\frac{igE_{i,i+1}t}{\eta}\right] \quad \text{g = coupling constant,} \quad i = 1 \ldots n-1$$

This set of gates is universal on each individual (n,J) DFS! (and not more!)

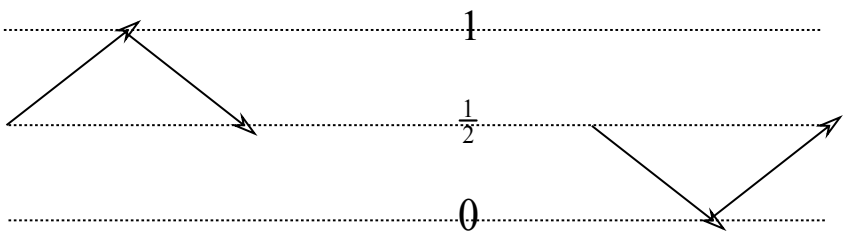
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n=3: the states are formed from combining a triplet with a spin-$\frac{1}{2}$ state or or by combining a singlet with a spin-$\frac{1}{2}$ state. Triplet is invarient under exchange and singlet flips sign under exchange. Thus

$E_{12} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ on the logical qubits.

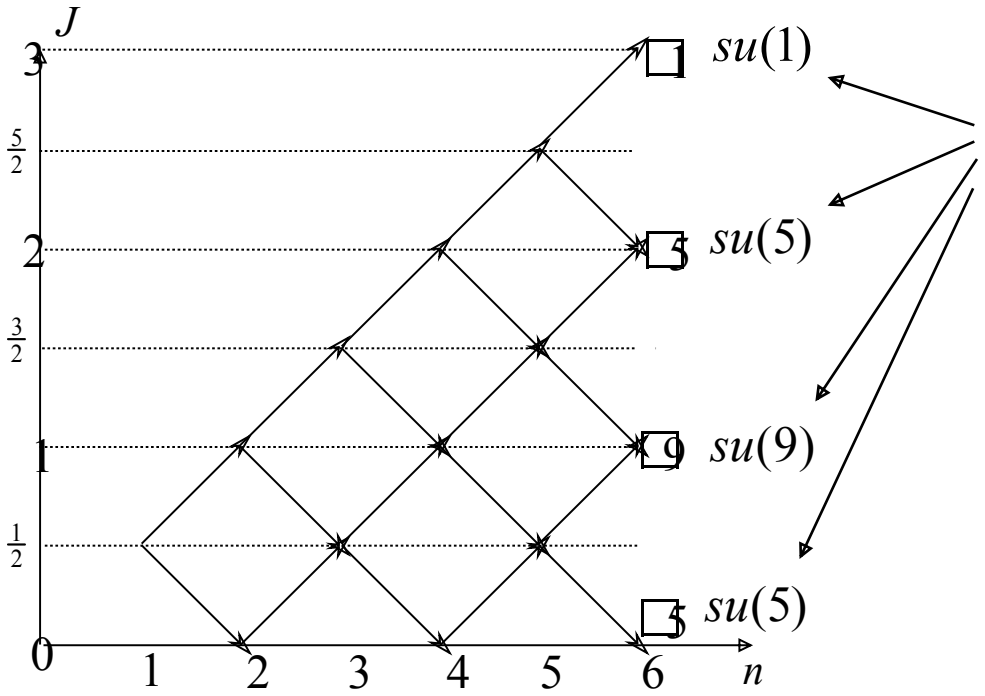
more details tell us that

$E_{23} \rightarrow \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ on the logical qubits.

$|0_L\rangle \equiv$ 

$\equiv |1_L\rangle$

Exchange's Universality



Can perform su(d)'s on each of the DFSs

encode information into the DFSs and the EXCHANGE interaction can be used to perform any su(d) on each of the DFSs (for a given n) independently!

su(d) means we have a d dimensional Hilbert space (encoded) and we can perform all unitary manipulations on this d dimensional Hilbert space!

Trick to proof: pathway in one-to-one correspondence with degeneracy

fault-tolerant quantum computation on collective DFSs:

Bacon, Kempe, Lidar, Whaley, PRL **85**, 1758 (2000)

Kempe, Bacon, Lidar, Whaley, to appear in PRA (2001)

Quantum Information is Fungible

Fungible *adj.* (New Latin *fungibilis*, from Latin *fungi* to perform)
1. being of such a nature that one part or quantity may be replaced by another equal part or quantity in the satisfaction of an obligation.
2. interchangable.

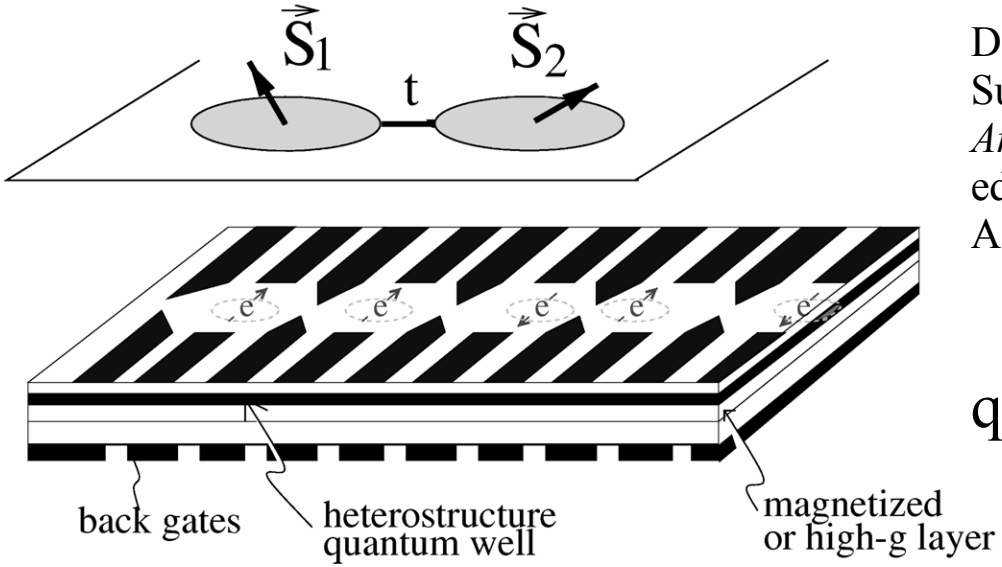
Value of classical information is independent of the medium:
- spin of millions of electrons
- presence or absence of a chad[1]
- etc.

Classical information is fungible. (can be encoded)

Quantum information is fungible. (can be encoded)

[1] U.S. Supreme Court case no. 00-949, George W. Bush vs. Albert Gore Jr.

Spintronic Quantum Computer



D.P. DiVincenzo, G. Burkard, D. Loss, E.V. Sukhorukov, in *Quantum Mesoscopic Phenomena And Mesoscopic Devices In Microelectronics,* eds. I.O. Kulik and R. Elliatioglu (NATO Advanced Study Institute, Turkey) 1999

qubit = spin of single electron

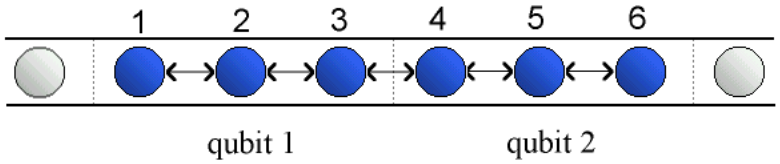
exchange interaction between qubits used for two-qubit interaction: tuned by raising and lowering barrier between dots to change overlap.

coupling strength: on ≈0.1 meV, off many orders of magnitude smaller

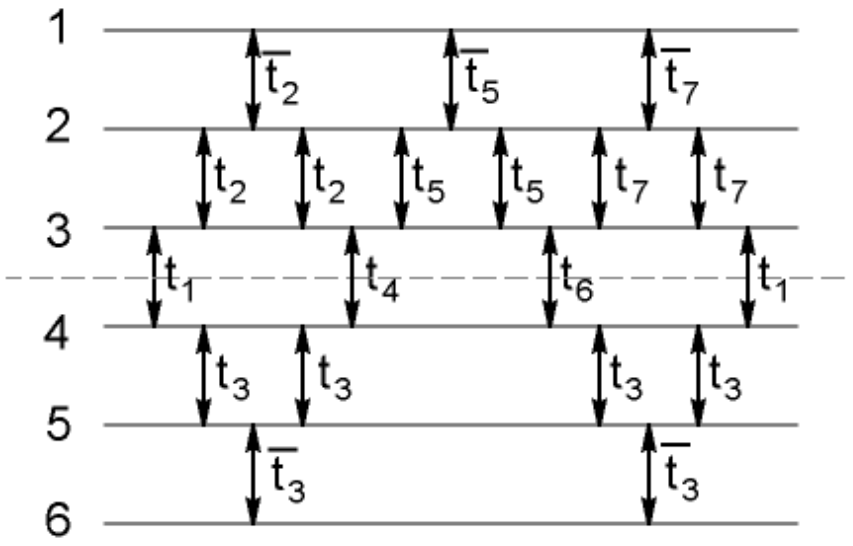
~~single qubit gates need to supplement exchange to make "fully~~ universal"

single qubit gates = huge demands on g-factor engineering, strong inhomogeneous magnetic fields, slow microwave manipulations.

Fungible Nature = Encoded Universality

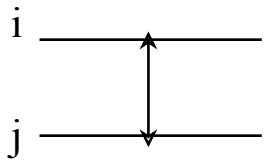


Nearest neighbor exchange coupling



t_1=0.410899(2) t_5=0.414720(10)
t_2=0.207110(20) t_6=0.147654(12)
t_3=0.2775258(12) t_7=0.813126(12)
t_4=0.640505(8) $\tan(\pi t)\tan(\pi \bar{t}) = -2$

exchange gate



$= \exp\left(iJE_{i,j}t\right)$

DiVincenzo, Bacon, Kempe, Burkard, Whaley, *Nature* **408**, 339 (2000)

Tradeoffs

factor of 3 in space

factor of ~ 10 time

What is Necessary to Quantum Compute?



Hey I found this new quantum system with these crazy interactions. Can I quantum compute?

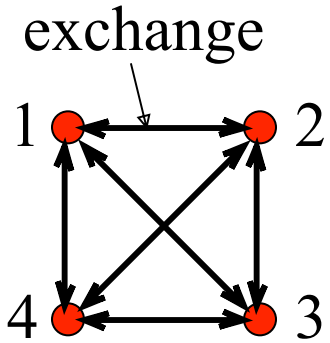
- **"Encoded universality"** changes the rules for physical implementations of quantum computers
- **Exchange is universal with a bit of encoding**
- **Other interactions** besides exchange have similar properties (simple application of representation theory of Lie algebras: *take it to your local particle theorist for a quick answer!*)



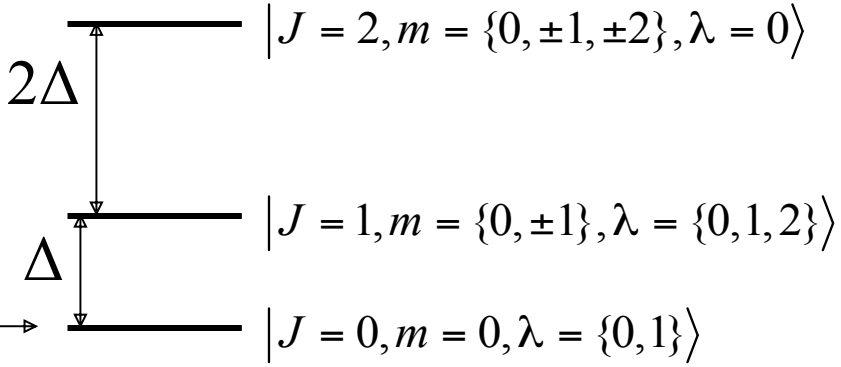
Controlled-Not and single qubit gates are not the only way to build a quantum computer!



The Future: Supercoherence



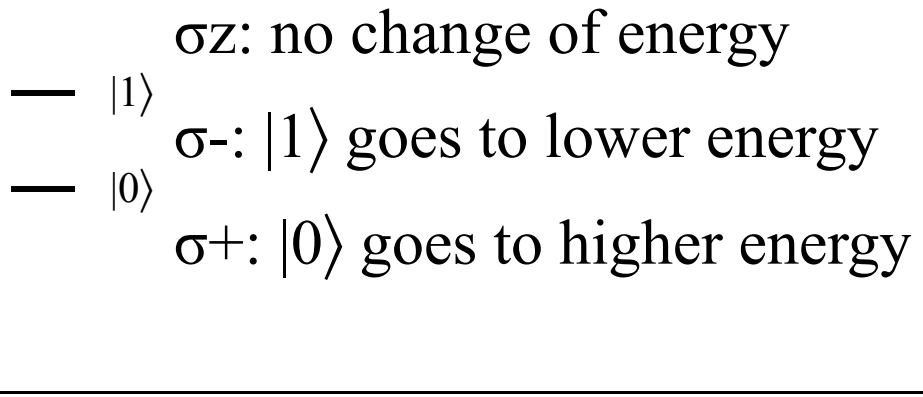
n=4

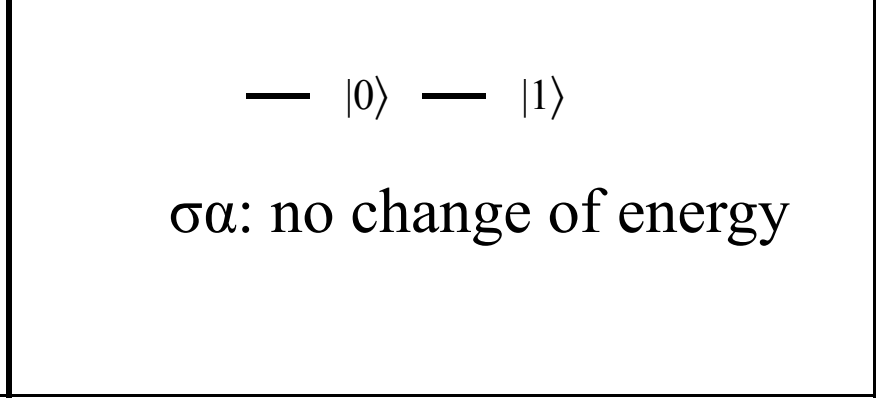


degenerate ground state

All single qubit operators must take J=0 to J=1 states (selection rule)

Low bath temperatures, single qubit errors must heat system





Conclusion

- Decoherence-free subsystems use symmetry of system-bath coupling to avoid decoherence
- Use symmetry first: concatenate DFSs into quantum error correction codes
- DFSs robust to perturbation
- Fault-tolerant quantum computation on collective DFSs
- Exchange has "encoded universality"
- Fungible nature of quantum information impacts engineering of quantum computer architectures